

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q15-0551

Lone Tree 138 kV  
August 22, 2017

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0551. The Interconnection Customer proposes a 126.3 MW solar photovoltaic facility interconnecting at the Lone Tree 138 kV substation in the Virginia area. The request was received on December 14, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 226 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,898,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                              | Contingency                                     | Loading (%) | Rating (MVA) | DFAX (%) |
|-------------------------------------------------|-------------------------------------------------|-------------|--------------|----------|
| Osceola - Michigan Power Lp 138 kV              | Lone Tree - Craven County Wood Energy Lp 138 kV | 122.5       | 355          | 100.0    |
| Lone Tree - Oscola 138 kV                       | Craven County Wood Energy Lp - Lone Tree 138 kV | 121.9       | 361          | 100.0    |
| Haysi Substation - Tom Watkins 138 kV Line      | Atoka - Catawba 345 kV                          | 109.9       | 266          | 5.5      |
| Crandall Substation - Carodean 138 kV           | Crandall - Dayton 345 kV line                   | 104.7       | 236          | 5.2      |
| Grottoes - Weyers Cave 138 kV                   | Crandall - Dayton 345 kV line                   | 102.9       | 352          | 5.0      |
| Craven County Wood Energy Lp - Lowmoor 138 kV L | Lone Tree - Osceola 138 kV line                 | 100.6       | 433          | 100.0    |
| Craven County Wood Energy Lp - Lone Tree 138 kV | Osceola - Lone Tree 138 kV                      | 100.5       | 438          | 100.0    |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                        | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Osceola - Michigan Power Lp 138 kV              | Rebuild line with higher-capacity conductor | \$9,322,000   | 28% (\$2,597,000)     |
| Lone Tree - Osceola 138 kV                      | Rebuild line with higher-capacity conductor | \$4,764,000   | 28% (\$1,327,000)     |
| Haysi Substation - Tom Watkins 138 kV           | Reconductor line                            | \$9,156,000   | 7% (\$644,000)        |
| Carodean - Crandall 138 kV                      | Reconductor line                            | \$1,313,000   | 8% (\$108,000)        |
| Grottoes - Weyers Cave 138 kV                   | Reconductor line                            | \$2,395,000   | 6% (\$150,000)        |
| Craven County Wood Energy Lp - Lowmoor 138 kV   | Reconductor line                            | \$3,280,000   | 28% (\$914,000)       |
| Craven County Wood Energy Lp - Lone Tree 138 kV | Reconductor line                            | \$4,158,000   | 28% (\$1,158,000)     |
|                                                 | Total allocated to project                  |               | \$6,898,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.